# Predicting "It Bags" Using Resale Market and Public Attention Signal

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## Predicting

Luxury handbags show highly uneven resale behavior: most depreciate while a small subset become iconic **"It Bags"** that retain or exceed their original retail price. Identifying these breakout products early is valuable for resellers, collectors, and secondary marketplaces allocating inventory under uncertainty.

**We study whether observable early signals from resale listings and public attention can predict future resale dominance.**

Our dataset contains **364 handbag models** with features derived from resale market activity and Google Trends search interest. We train several machine learning models to predict whether a handbag will fall into the **top 20% of resale performance**.

## Data and Features

Our dataset contains **364 luxury handbag models** collected from from the resale platform Rebag (Rebag 2026). For each model, we construct **25 model-level features** capturing resale market activity, pricing signals, product condition, and public attention. The raw inputs include listing statistics (resale prices, views, favorites, listing counts) and Google Trends search volumes. We derive additional features such as log-transformed engagement metrics and ratios (e.g., favorites-to-views) to better represent demand intensity

The binary label `it_bag` identifies models whose **median resale premium falls within the top 20%** of all models. This yields an **imbalanced classification problem**, class distribution is reported in table 1.

| Dataset Split | Not-It-Bag | It-Bag | Total |
|---|---|---|---|
| Full Dataset | 262 | 66 | 328 |
| Train + Validation | 209 | 53 | 262 |
| Holdout | 53 | 13 | 66 |

Table 1. Class distribution full dataset, train/ validation set and holdout set.

## Models

### Logistic Regression (L1 / L2)

Parameters are learned by minimizing the regularized logistic loss

$$\mathcal{L}(w,b) = -\frac{1}{n}\sum_{i=1}^{n}\Big[y_i \log \sigma(w^\top x_i + b) + (1-y_i)\log(1-\sigma(w^\top x_i + b))\Big] + R(w),$$

where $w \in \mathbb{R}^d$ is the weight vector, $b$ is a bias term, $\sigma(z) = \frac{1}{1+e^{-z}}$ is the logistic sigmoid function and $R(w)$ is a regularization term (L1- or L2-norm) (Hastie, Tibshirani, and Friedman 2009).

### Tree-based methods

**Random Forest** (Breiman 2001) averages the predictions of an ensemble of decision trees trained on bootstrap samples. **Gradient Boosted Trees** are build sequentially, each tree is trained to correct the residual errors of the previous ensemble. Trees are split such that Gini impurity is minimized

$$G = \rho_{0,m}(1-\rho_{0,m}) + \rho_{1,m}(1-\rho_{1,m}),$$

where $\rho_{k,m}$ is the proportion of observations of class $k = \{0,1\}$ in node $m$.

## Results

Because the dataset is imbalanced, we use the area under the precision–recall curve (AUPRC) as the primary metric for model comparison. Perfomance is reported in table 2.

| Model | Validation AUPRC (n=262) | Holdout AUPRC (n=66) |
|---|---|---|
| **Logistic Regression (L1)** | **0.752** | 0.879 |
| Histogram Gradient Boosting | 0.748 | 0.920 |
| Random Forest | 0.746 | 0.845 |
| Logistic Regression (L2) | 0.731 | 0.915 |

Table 2. Model performance using cross-validated AUPRC and holdout AUPRC.

Both linear and nonlinear models achieve strong predictive performance, suggesting that **resale market signals contain substantial information about breakout handbag success**.

L1-regularized logistic regression performs as well as more complex ensemble models, indicating that a relatively small subset of features captures most of the predictive signal.

Figure 1 shows the precision–recall curve for the L1-regularized logistic regression model, which is the best-performing model based on cross-validated AUPRC.

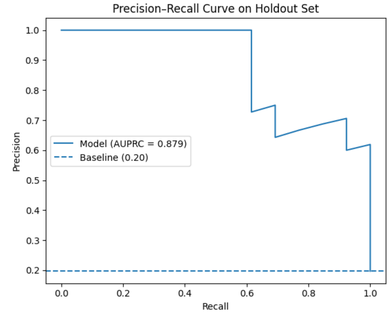


Figure 1. Precision-Recall Curve for L1-regularized logistic regression

The **confusion matrix** of the L1-regularized regression model on the holdout set is:

$$\begin{bmatrix} 51 & 2 \\ 5 & 8 \end{bmatrix}$$

The model identifies 8 of 13 breakout handbags with only 2 false positives. This indicates that the classifier is relatively conservative in predicting breakout products, **prioritizing precision over recall**.

**Class imbalance** motivates the use of Synthetic Minority Oversampling Technique (**SMOTE**). In our experiments, the performance of SMOTE-augmented models remains comparable or slightly lower than the class-weighted baseline. This indicates that the dataset already contains **sufficient signal for learning the minority class**.

We also conduct a **temporal validation experiment**: models are trained on earlier handbag observations and evaluated on later entries. Performance remains comparable to the random holdout evaluation, suggesting that the **predictive signals remain stable across time.**

## Feature Importance and Discussion

**SHAP (SHapley Additive exPlanations)** (Lundberg and Lee 2017) shows that resale pricing signals, engagement metrics, and public attention indicators are among the most influential drivers of predictions. Figure 2 shows SHAP values for a datapoint on the holdout set that was correctly classified as "It bag" by the L1-regularized logistic regression model. For this instance, the estimated retail price holds the most explanatory power.

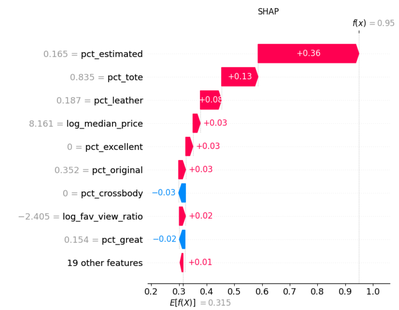


Figure 2. SHAP values for a correctly classified "It Bag" in the holdout set.

Furthermore, we conduct an **ablation study** by removing groups of features and retraining the model. Feature groups include engagement metrics, condition composition, product style and material composition, resale source indicators, and public attention signals (Google Trends). Model performance deteriorates most when Google Trends features or pricing signals are excluded, indicating that **public attention signals are the informative for predicting "It bag" status**.

## Future Work

Expand the dataset to include additional resale platforms and a larger set of handbag models, which would improve model stability and generalization.

Incorporate richer temporal signals such as early resale activity, social media attention, and time-series Google Trends features to better capture how demand evolves after product release.

Extend the framework toward a true forecasting setting that predicts breakout status using only information available shortly after a handbag's launch.

## References

Breiman, Leo (2001). "Random Forests". In: *Mach. Learn.* 45.1, pp. 5–32. ISSN: 0885-6125.

Hastie, Trevor, Robert Tibshirani, and Jerome Friedman (2009). *The Elements of Statistical Learning*. Springer.

Lundberg, Scott M and Su-In Lee (2017). "A Unified Approach to Interpreting Model Predictions". In: *Advances in Neural Information Processing Systems*. Ed. by I. Guyon et al. Vol. 30. Curran Associates, Inc.

Rebag (2026). *Rebag Luxury Resale Platform*. Accessed: 2026. URL: `https://www.rebag.com/`.